

L2GTX: From Local to Global Time Series Explanations

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Why Explainability Matters

Deep learning models achieve **impressive accuracy** in time series classification, but they operate as **opaque black boxes** .



We need to understand why the model makes a prediction, not just what it predicts.

Applications of Time Series Classification



Healthcare

ECG analysis, patient monitoring



Finance

Market prediction, risk assessment



Safety-Critical

Fault detection, anomaly monitoring

In high-stakes applications, explanations help validate predictions, build trust, and support decision-making.

Three Critical Gaps

1

Many model-agnostic XAI methods treat time steps as independent features, **overlooking temporal dependencies** [1,2].

2

Most time-series XAI methods provide local explanations, while class-wise global explanations **remain limited** [3-5].

3

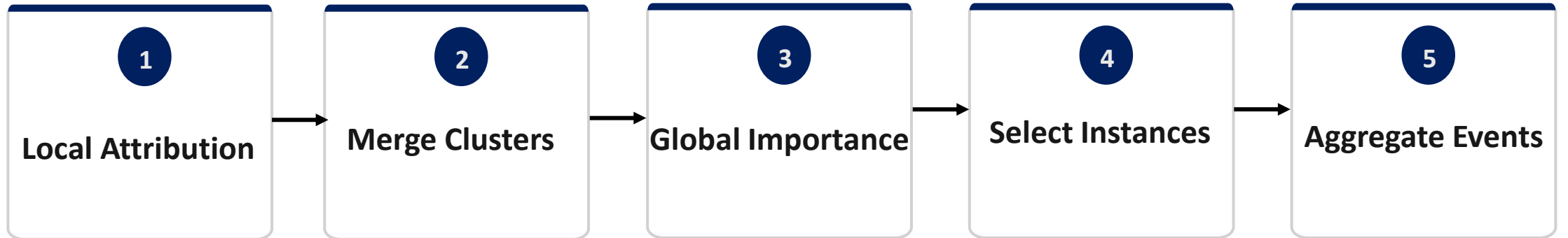
Existing global approaches are often **model-specific** or rely on **surrogate-rule extraction** [6-8].

?

How can we derive **class-wise global explanations** that preserve **temporal structure** while remaining **model-agnostic**?

L2GTX OVERVIEW

- Five-step process from local explanations to global insights.



Input

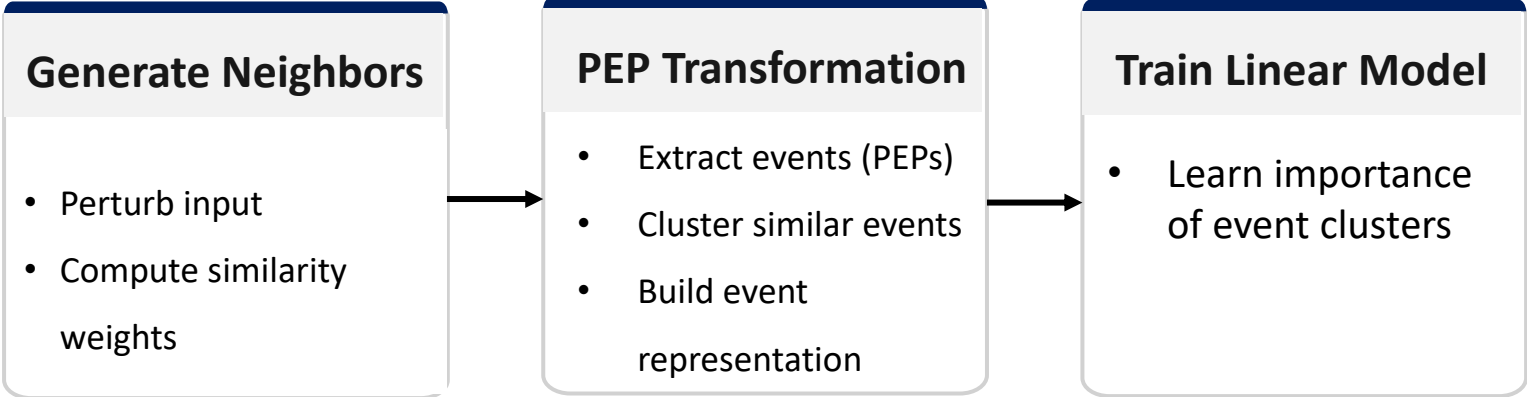
- Time series instances + black-box model f + Budget B

Output

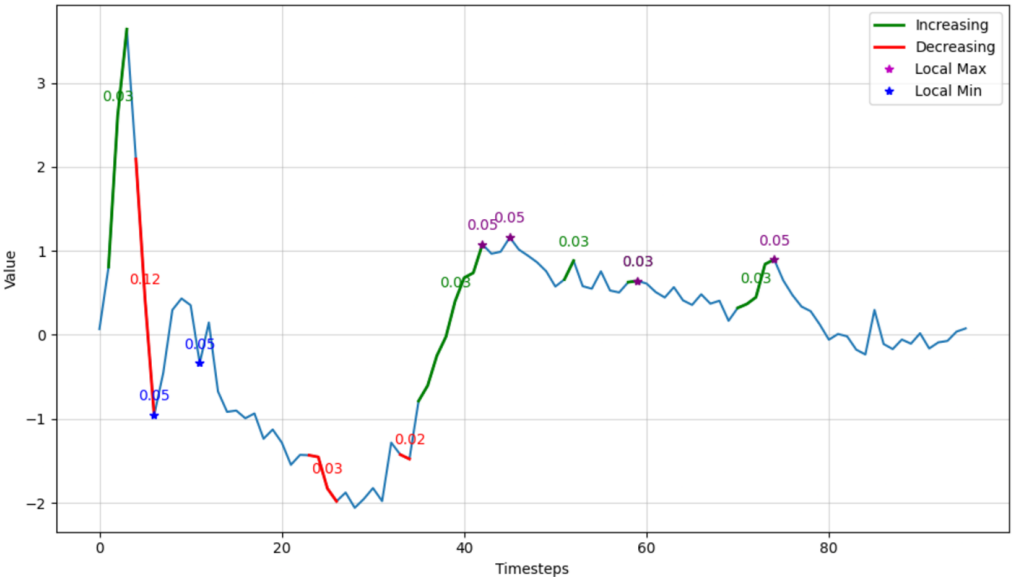
- Class-wise global explanations

Step 1: Local Explanations (LOMATCE)

- Generate local explanations using LOMATCE [9].



➤ **Parameterised Event Primitives (PEPs)** are domain-specific events characterised by parameters.



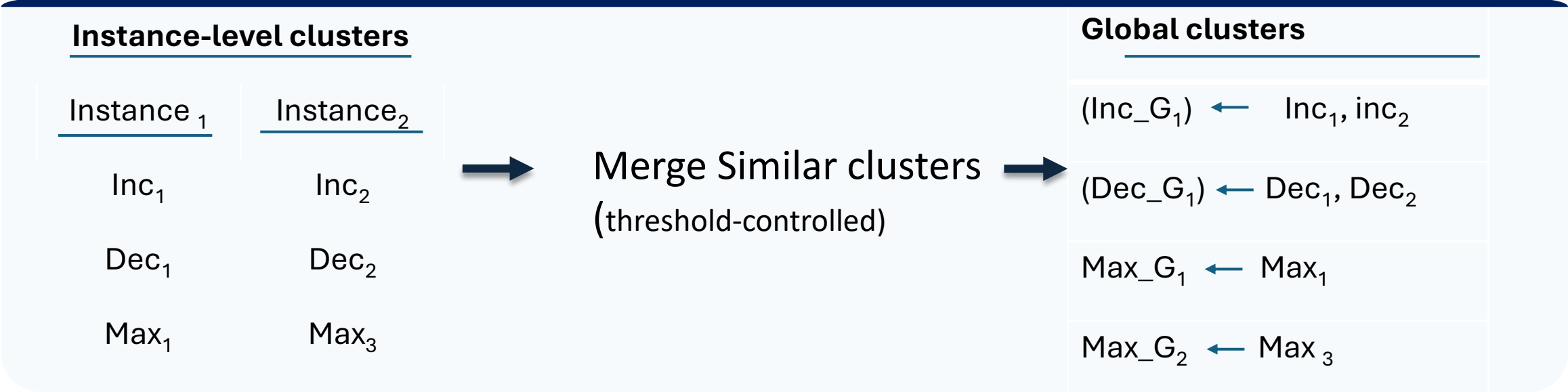
PEPs

Capture temporal patterns as human-understandable events:



Step 2: Merge Similar Clusters

- Top clusters with importance are obtained per instance.



- Similar clusters are merged using hierarchical clustering.
- Merging is controlled by a percentile-based threshold.

Step 3: Global Cluster Importance

- Aggregate across instances → Global importance per cluster.

	Inc_G ₁	Dec_G ₂	Max_G ₃
Inst ₁	0.8	0	0.3
Inst ₂	0	0.6	0.2
Inst ₃	0.7	0.5	0



$$I_j = \sqrt{\sum_{i=1}^N |M_{i,j}|}, \quad j = 1, \dots, G$$



Global Cluster Importance

N: instances

G: global clusters

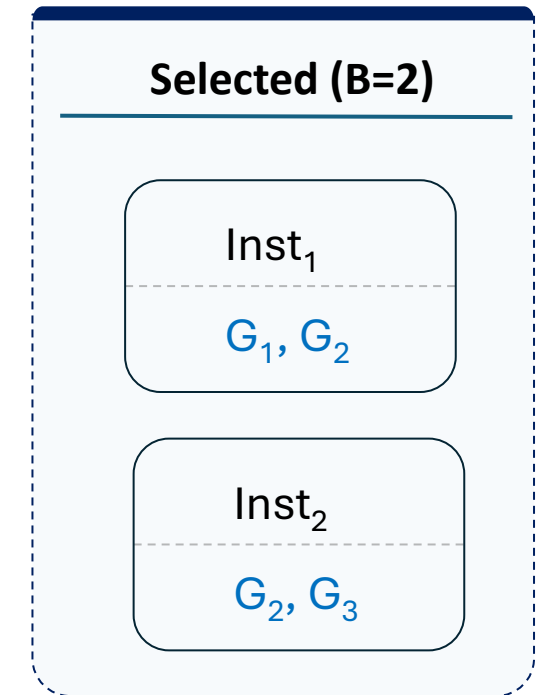
M_{i,j}: cluster importance *j* for instance *i*

Step 4: Select Representative Instances

- Greedily maximise additional coverage.

All Instances				
Inst ₁	Inst ₂	Inst ₃	Inst ₄	Inst ₅
G ₁ , G ₃	G ₂ , G ₃	G ₁	G ₃	G ₃ , G ₄

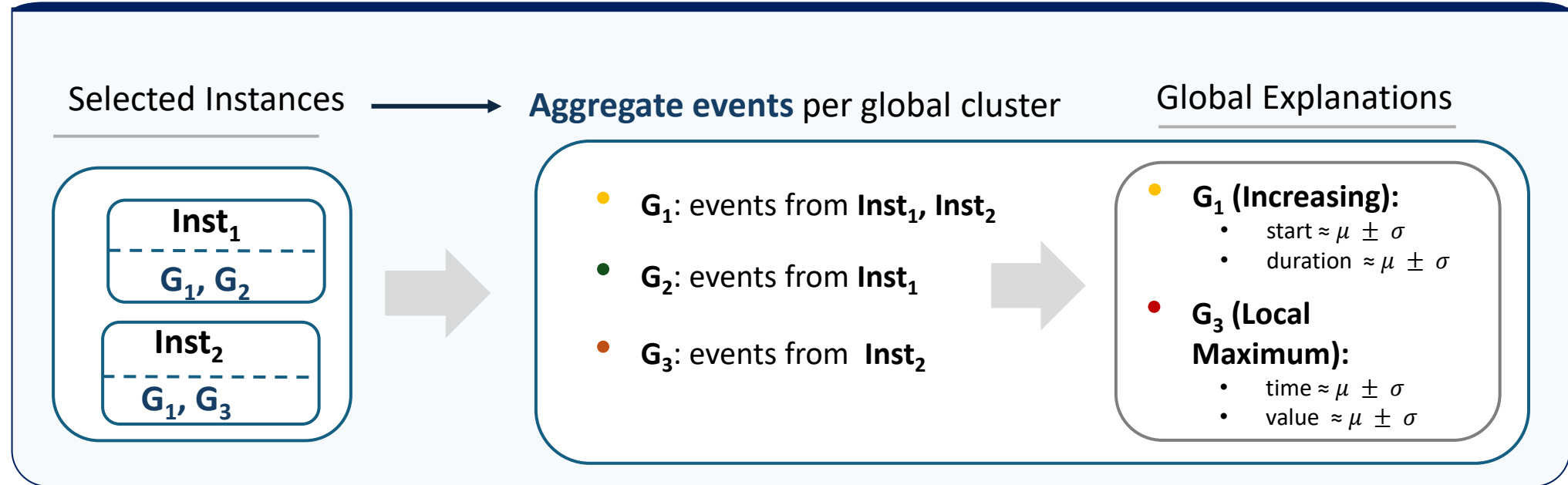
Submodular Pick [10]



Covered Clusters: G₁, G₂, G₃

- Maximise coverage of important clusters.
- Avoid selecting instances that explain the same clusters.
- Stop when all important clusters are covered (or budget B is reached).

Step 5: Aggregate Events Into Global Explanations



- Events assigned to the same global cluster are aggregated and summarised using *mean $\pm$ standard deviation* to produce class-wise explanations.

What L2GTX Produces

🔗 Class-Wise Global Explanations

Bar Chart Format

Importance scores of aggregated event clusters

Colour Coding

Different colours for event types (trends vs extrema)

Temporal Statistics

Mean $\pm$ std for timing, duration, magnitude

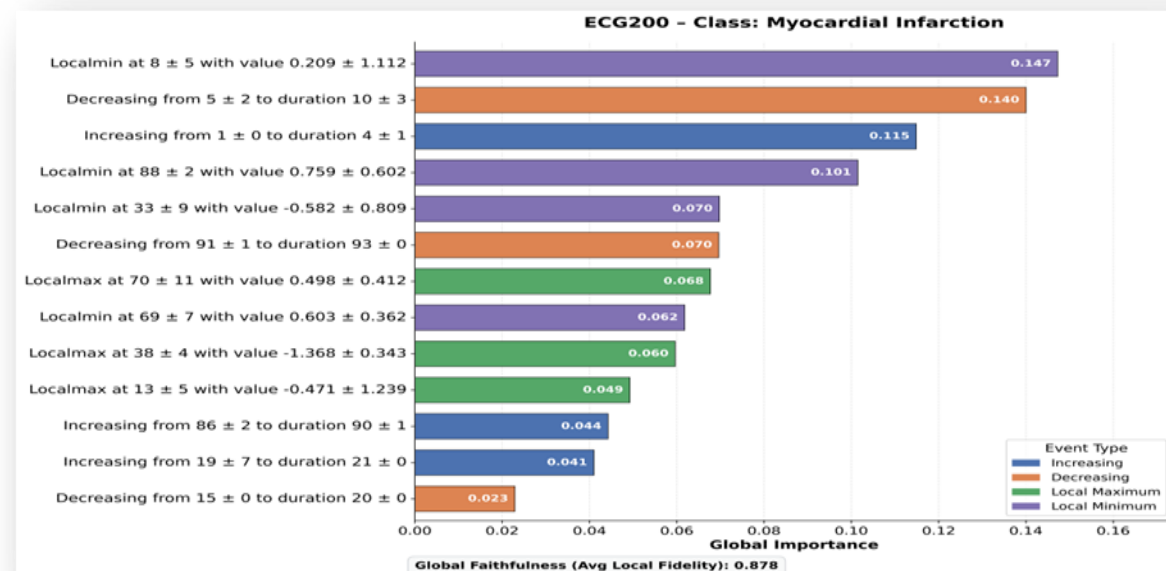
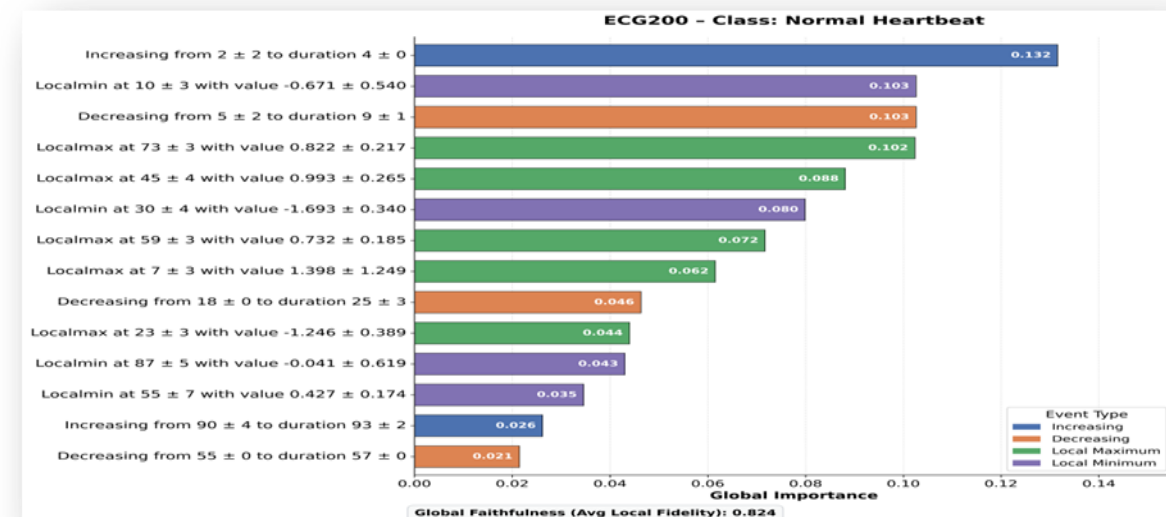
? What You Learn

WHAT patterns matter (trend/extrema)

WHERE they occur (temporal location)

HOW they manifest (duration, magnitude)

HOW MUCH they contribute (importance)



Black-Box Predictive Performance

- FCN and LSTM-FCN across six benchmark time series datasets.

**6 Benchmark Datasets**
Varying domains and difficulties

**2 Black-Box Models**
FCN • LSTM-FCN

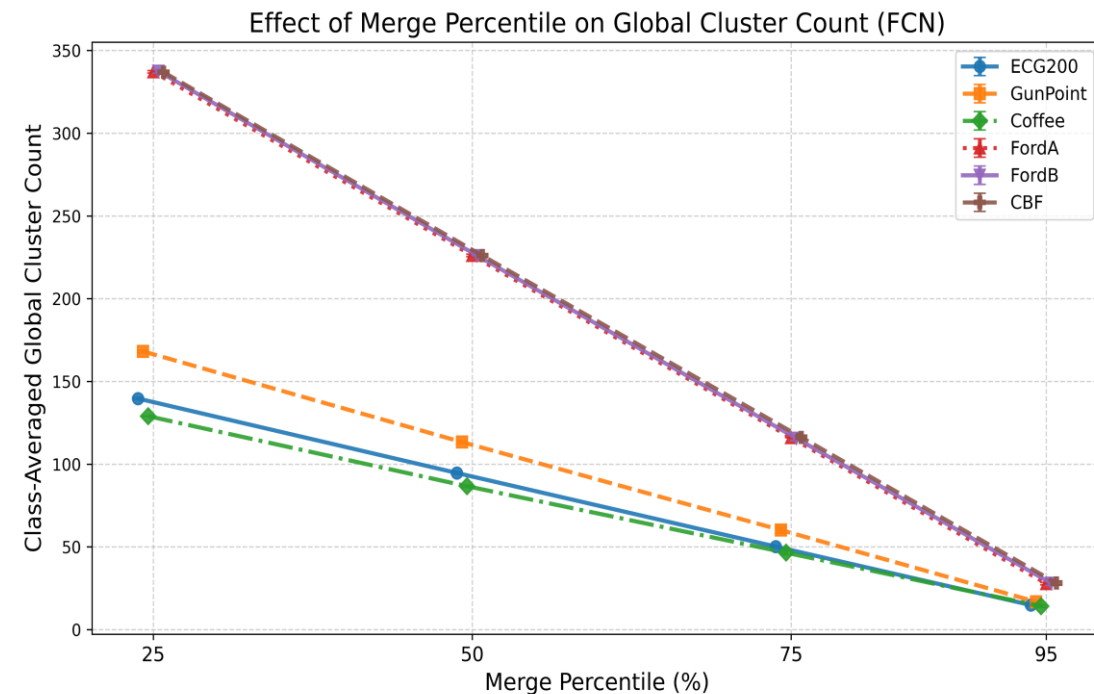
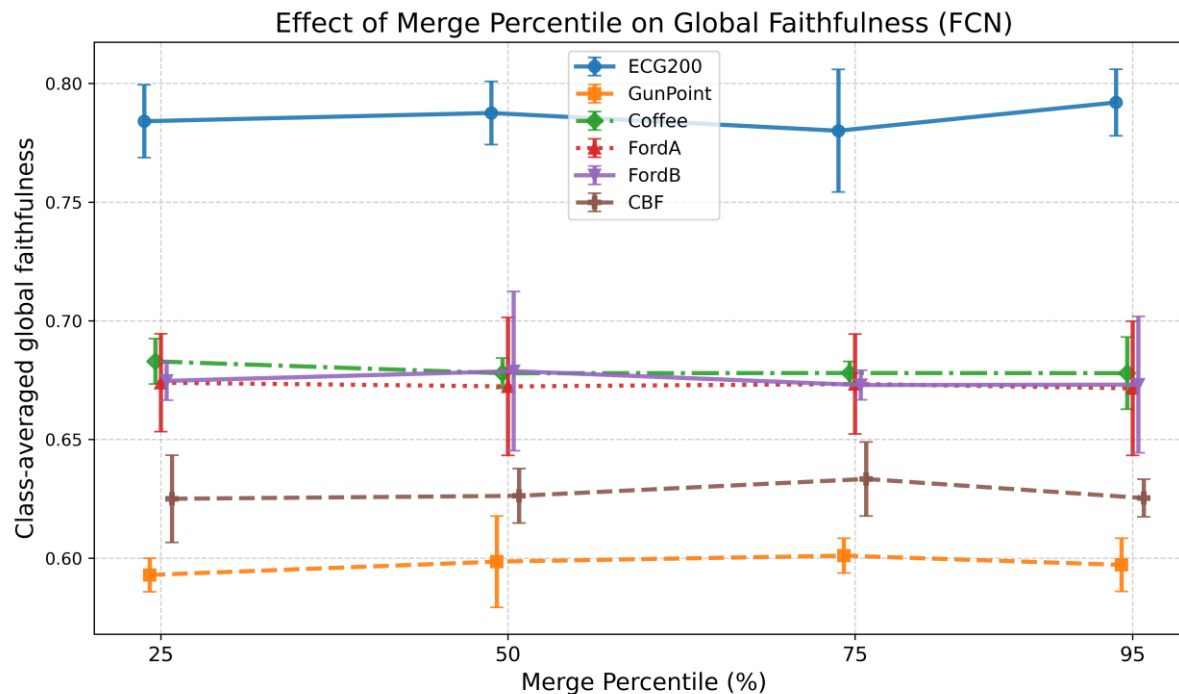
**100 Repeated Runs**
Random train/val/test splits
Mean accuracy ± 95% CI

Mean Test Accuracy (± 95% Confidence Interval)		
Dataset	FCN	LSTM_FCN
 ECG200	0.87 ± 0.03	0.86 ± 0.01
 GunPoint	0.99 ± 0.01	0.98 ± 0.01
 Coffee	1.00 ± 0.00	1.00 ± 0.00
 FordA	0.90 ± 0.02	0.88 ± 0.01
 FordB	0.88 ± 0.02	0.86 ± 0.01
 CBF	0.98 ± 0.01	1.00 ± 0.00

- Both models achieve high accuracy across all datasets, making them reliable targets for explanation.

Merge Percentile Sensitivity Analysis

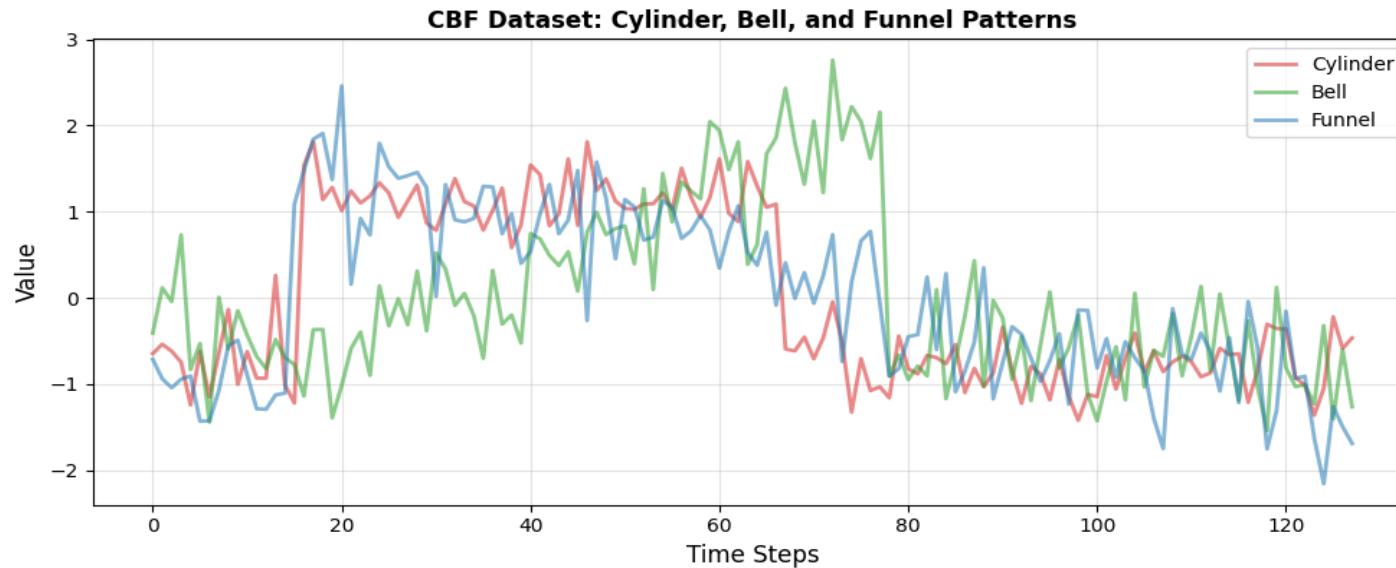
- Compact global explanations without loss of faithfulness.



- Global faithfulness remains largely unchanged across merge percentiles.
- **p = 95** provides the most compact explanations without sacrificing faithfulness.

- Increasing merge percentile progressively merges similar event clusters.
- The number of global clusters decreases substantially.

CBF Dataset – Shape-Based Time Series Classification



Cylinder

- Plateau bounded by two transitions.



Bell

- Gradual rise followed by a sharp decline.



Funnel

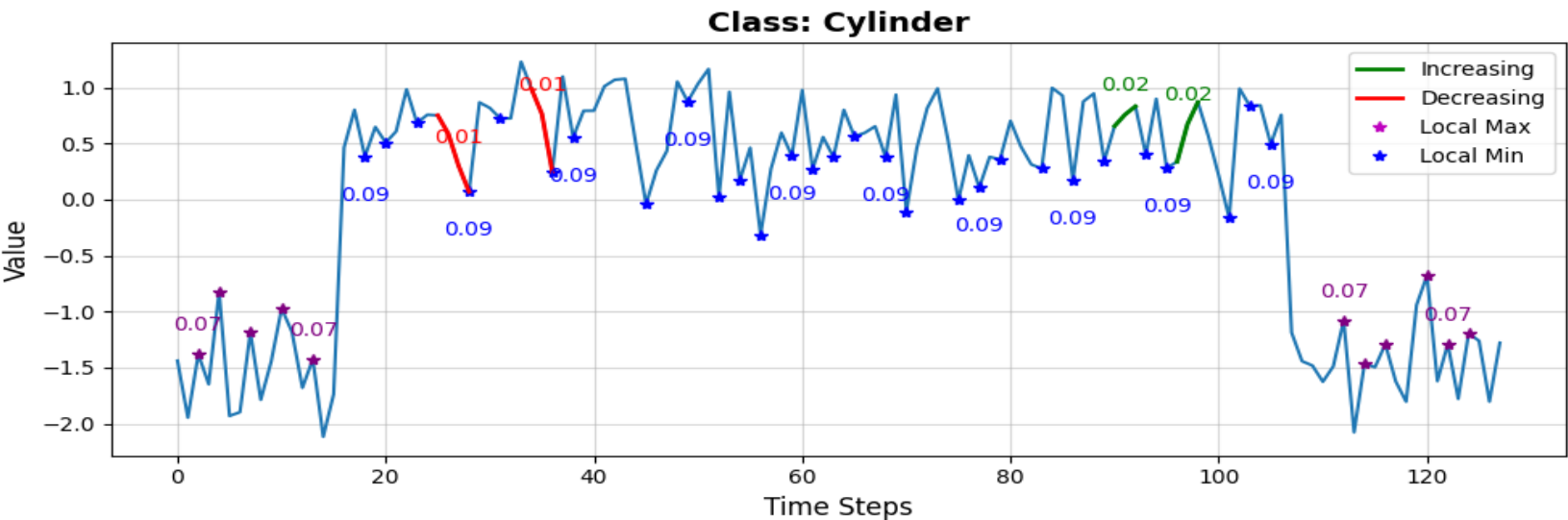
- Sharp rise followed by a gradual decline.

CBF

- Synthetic benchmark with known temporal patterns.
- Know class structure enables to evaluate whether explanations recover class-defining structures.

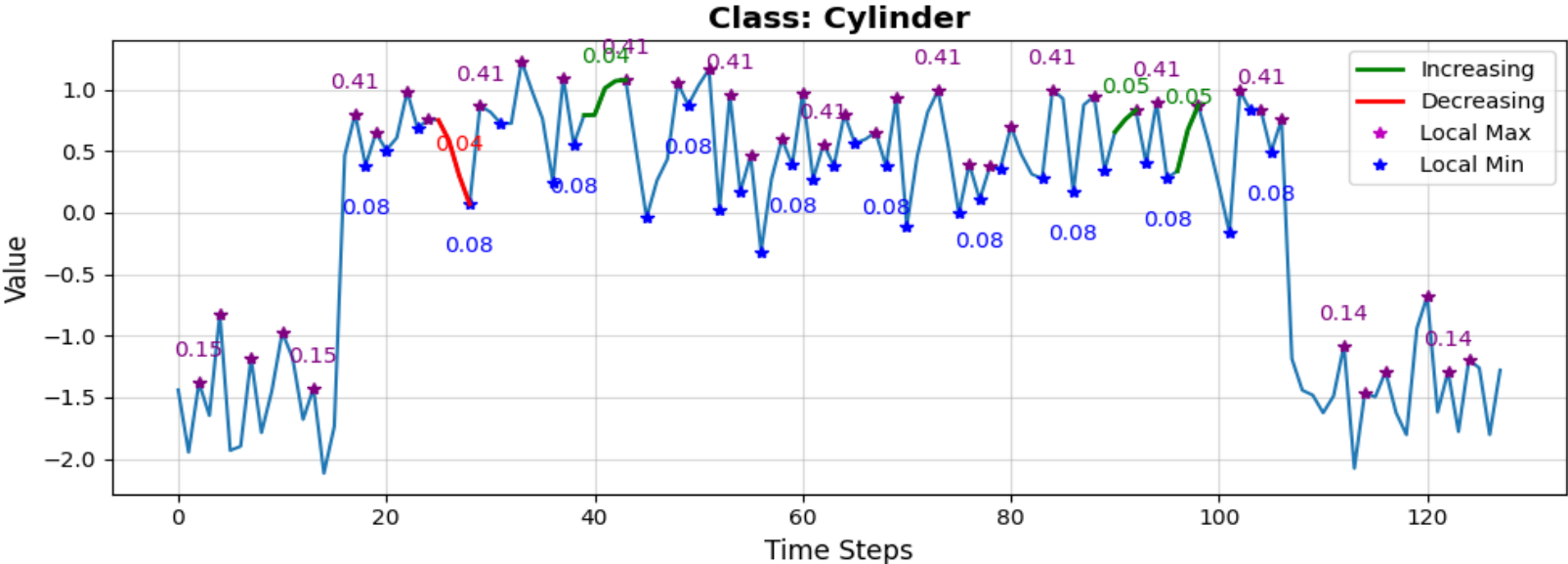
CBF: Local Explanations - Cylinder

FCN



- Important events occur around the plateau region and its boundaries.

LSTM-FCN



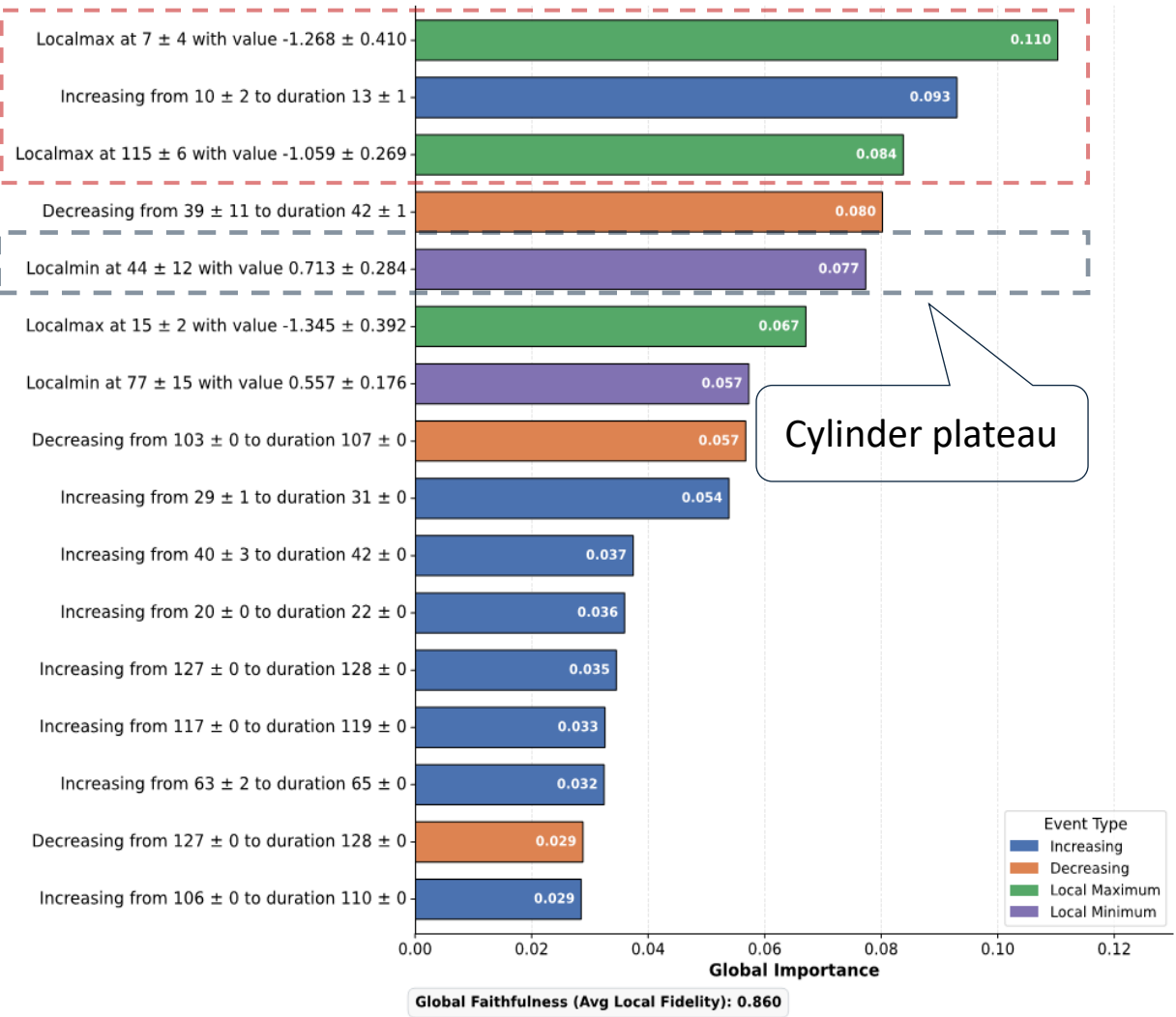
- Extrema and transitions contribute strongly to the prediction.

- The explanation is consistent with the expected Cylinder morphology.

CBF: L2GTX Global Explanations - Cylinder

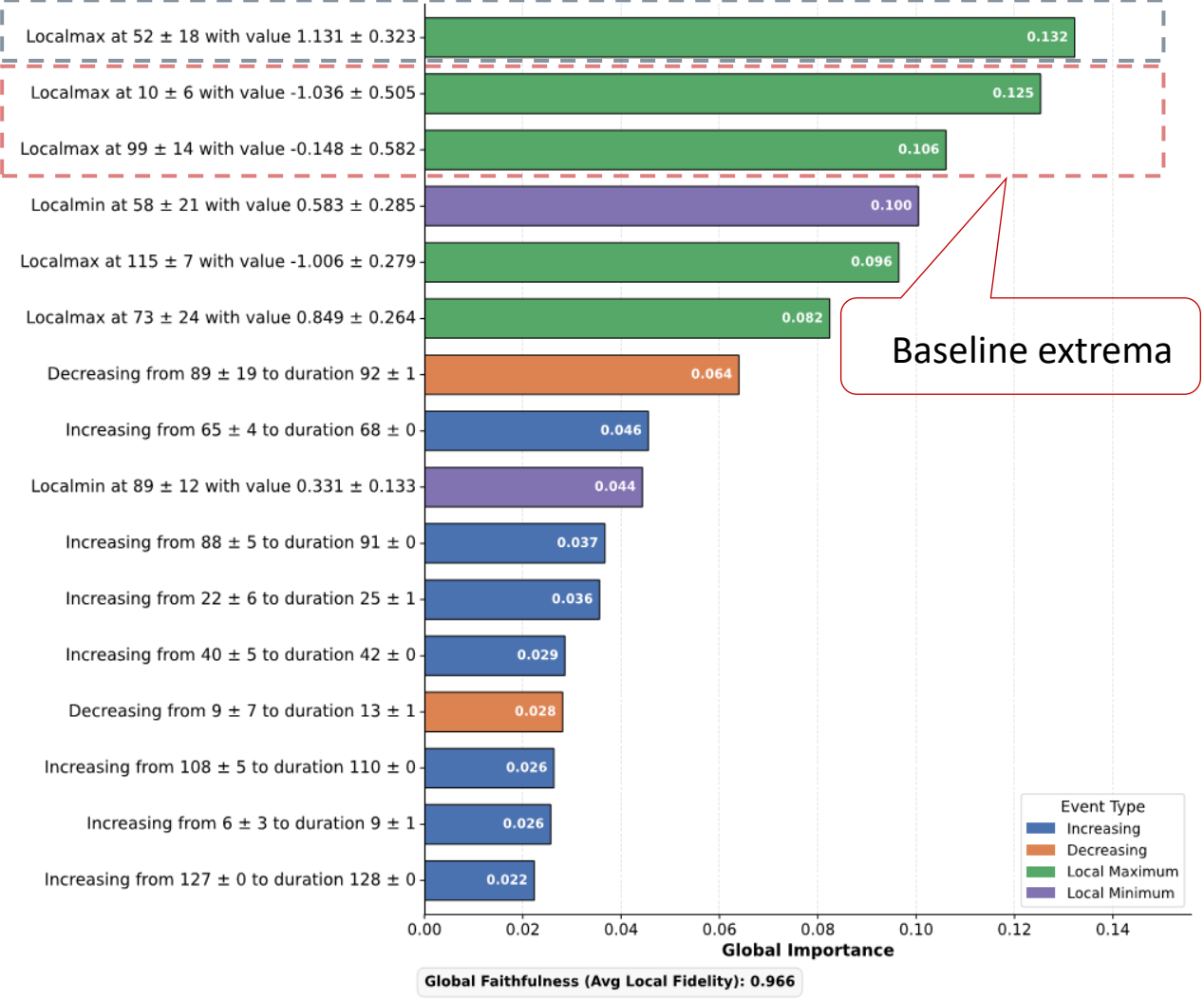
FCN

CBF - Class: Cylinder



LSTM-FCN

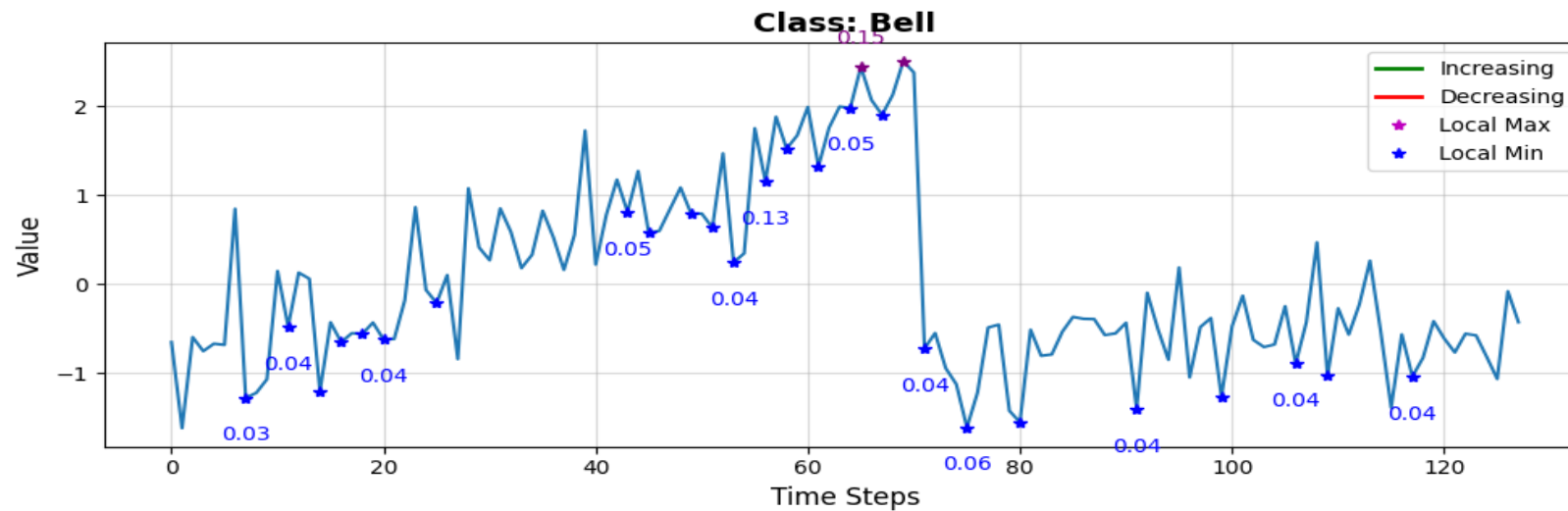
CBF - Class: Cylinder



- The most important events are concentrated around the plateau and surrounding baseline regions.

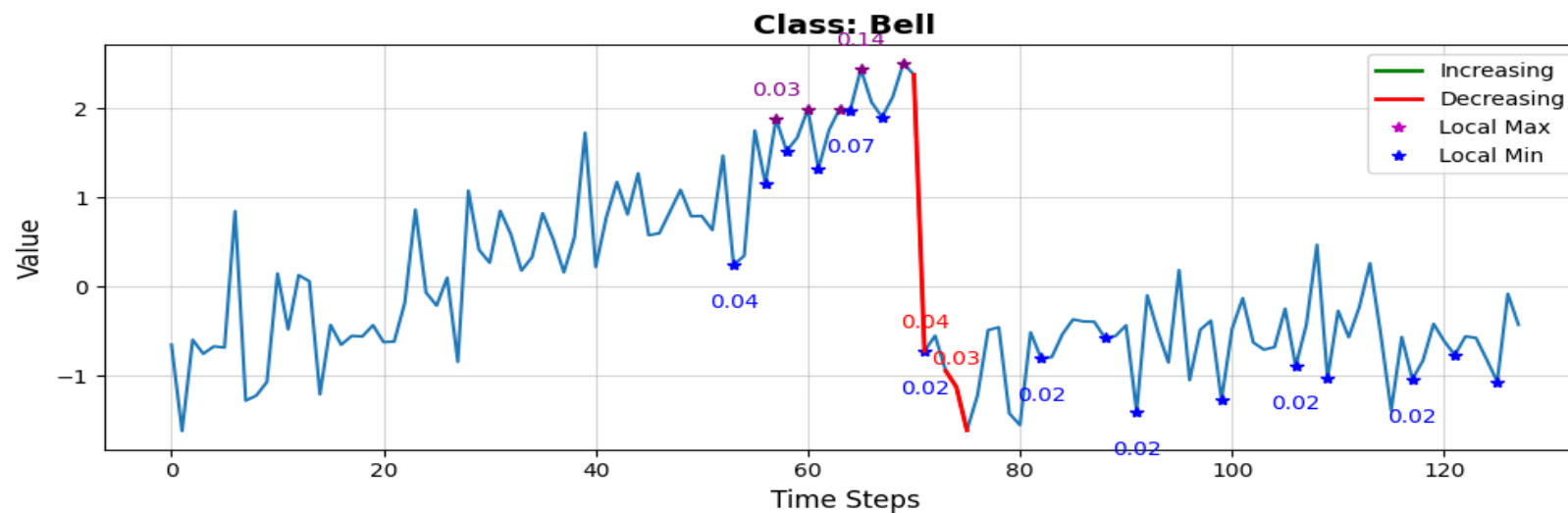
CBF: Local Explanations - Bell

FCN



- Important events are concentrated around the peak region.

LSTM-FCN

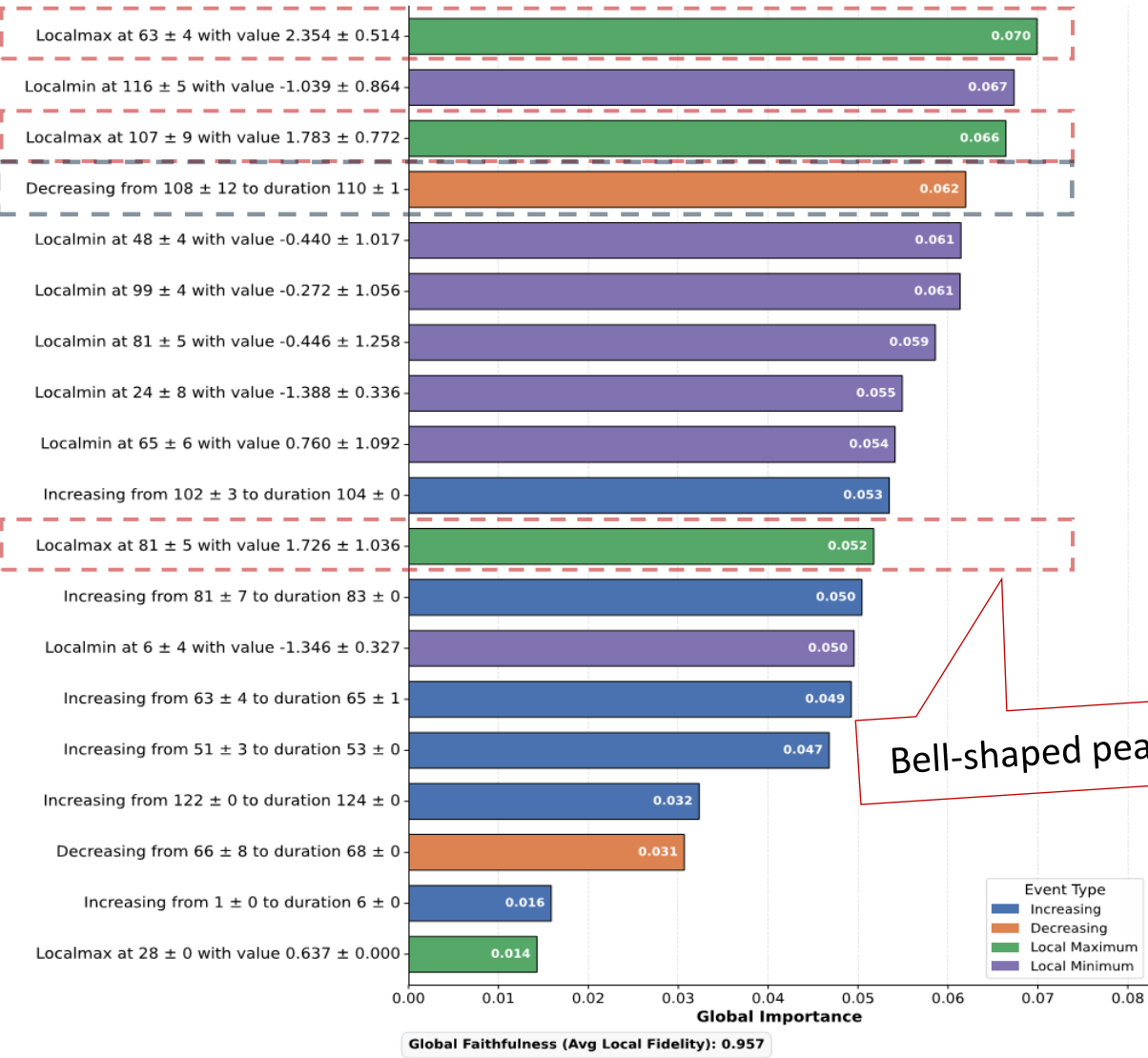


- Local extrema capture the characteristic Bell-shaped peak.

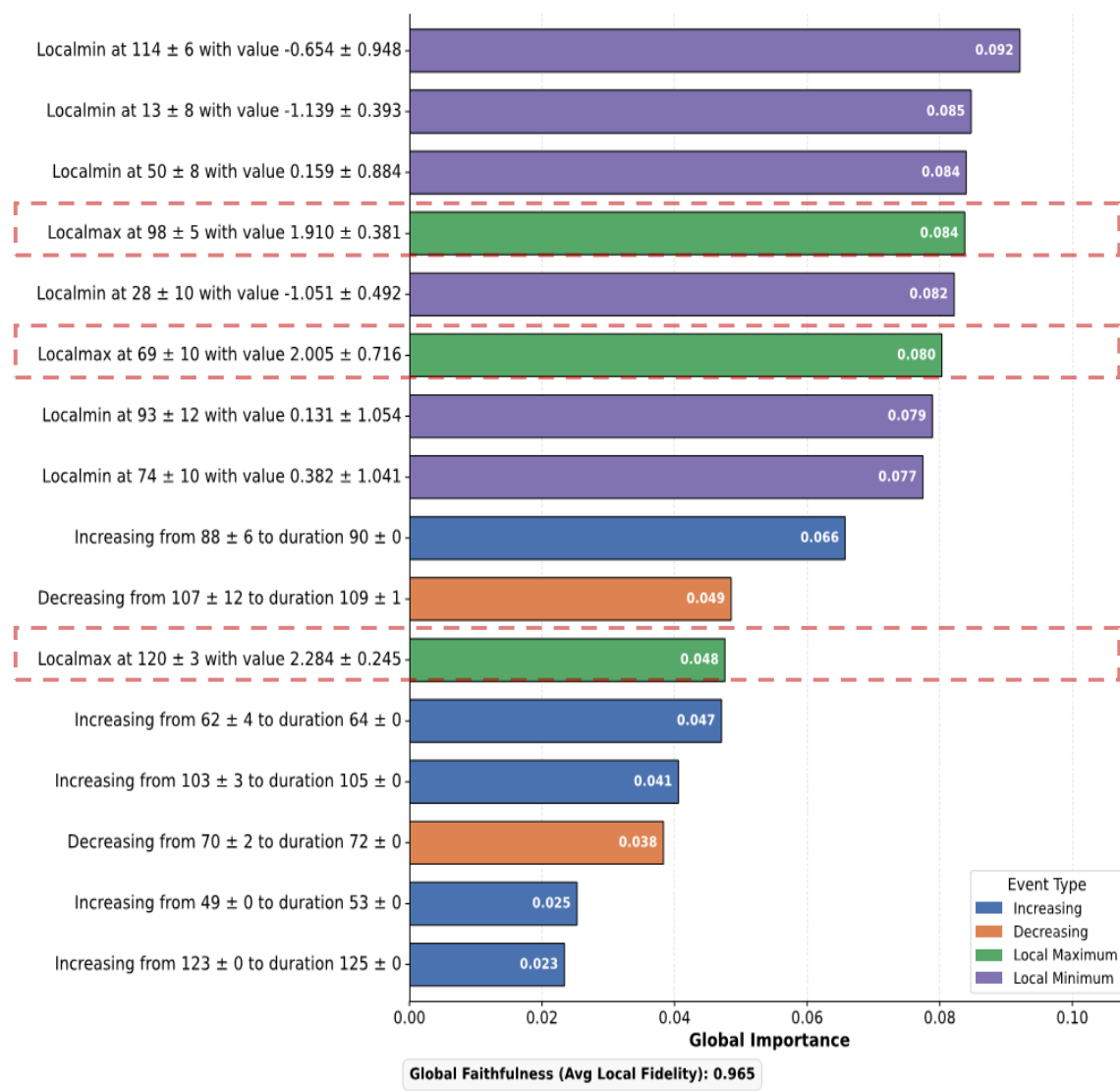
- Both models identify extrema around the peak region.

CBF: L2GTX Global Explanations - Bell

FCN



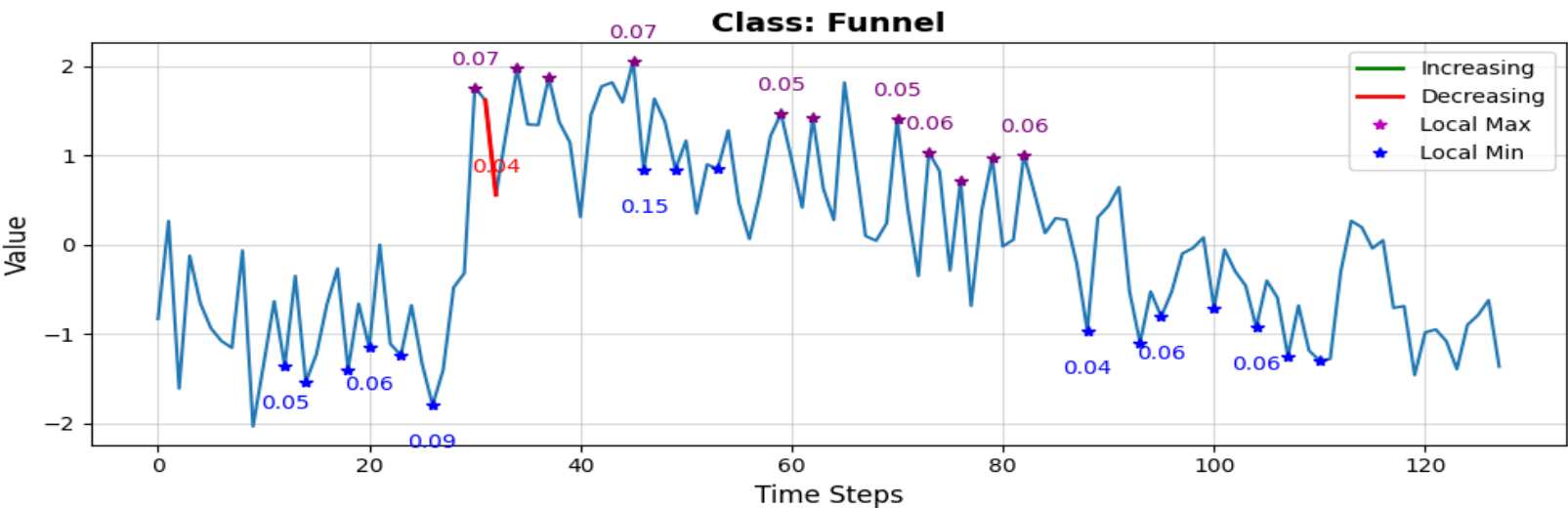
LSTM-FCN



- The most important events are concentrated around the characteristic Bell-shaped peak structure.

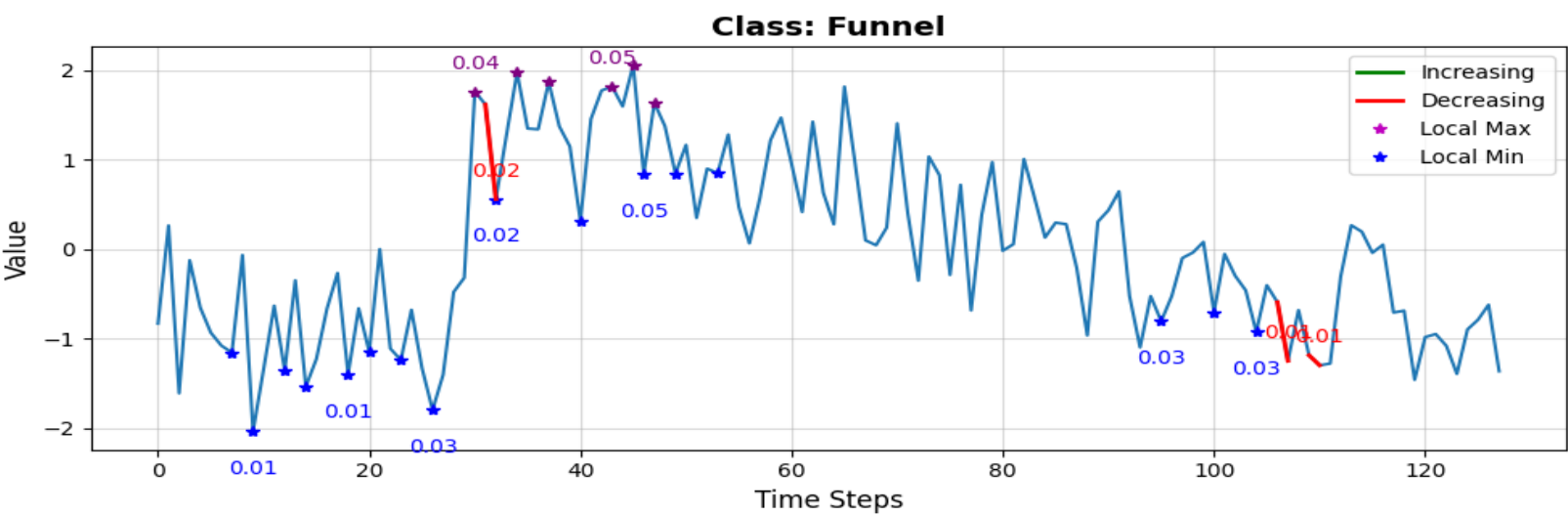
CBF: Local Explanations - Funnel

FCN



■ Important events occur around the transition into the high-value region.

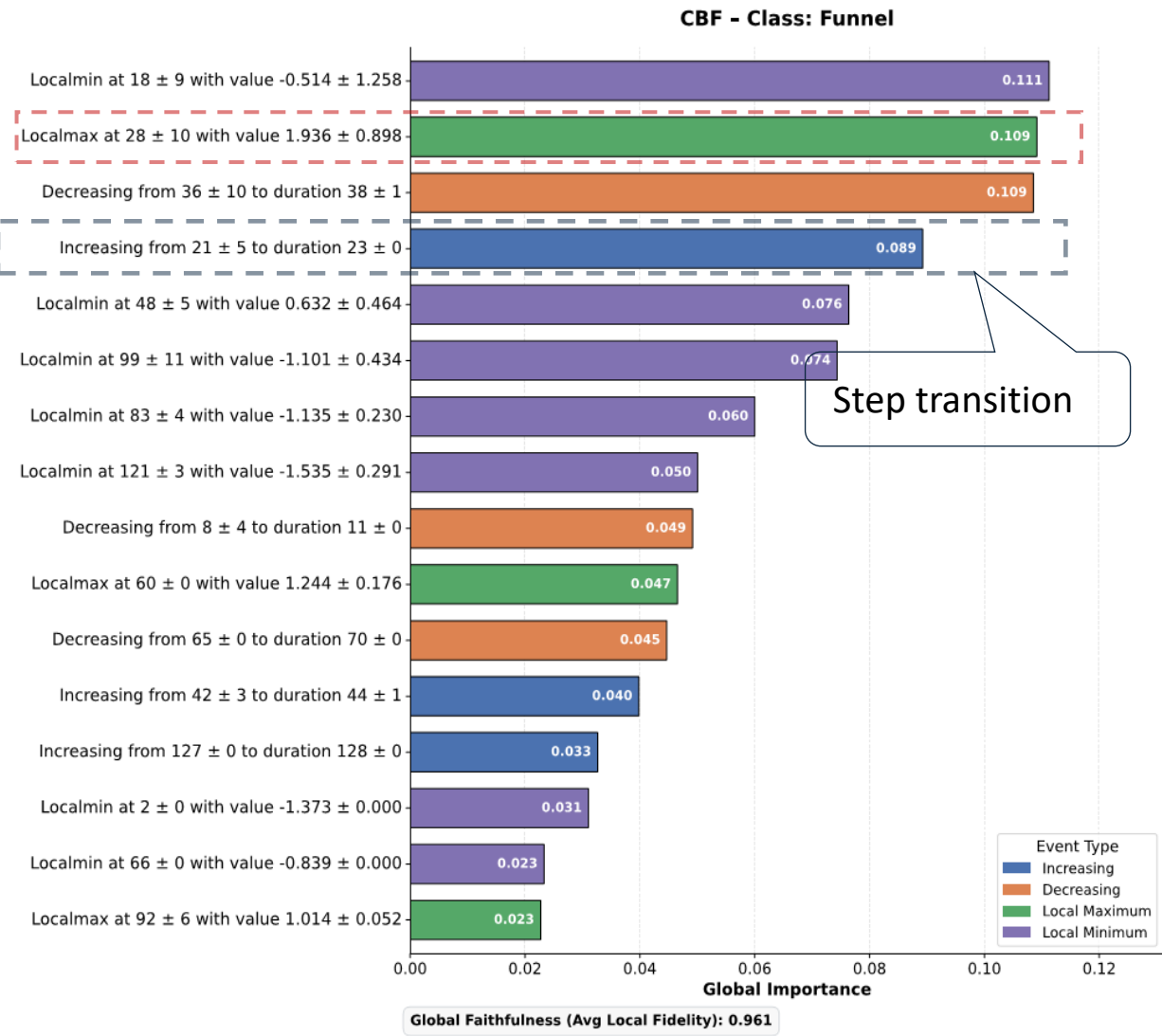
LSTM-FCN



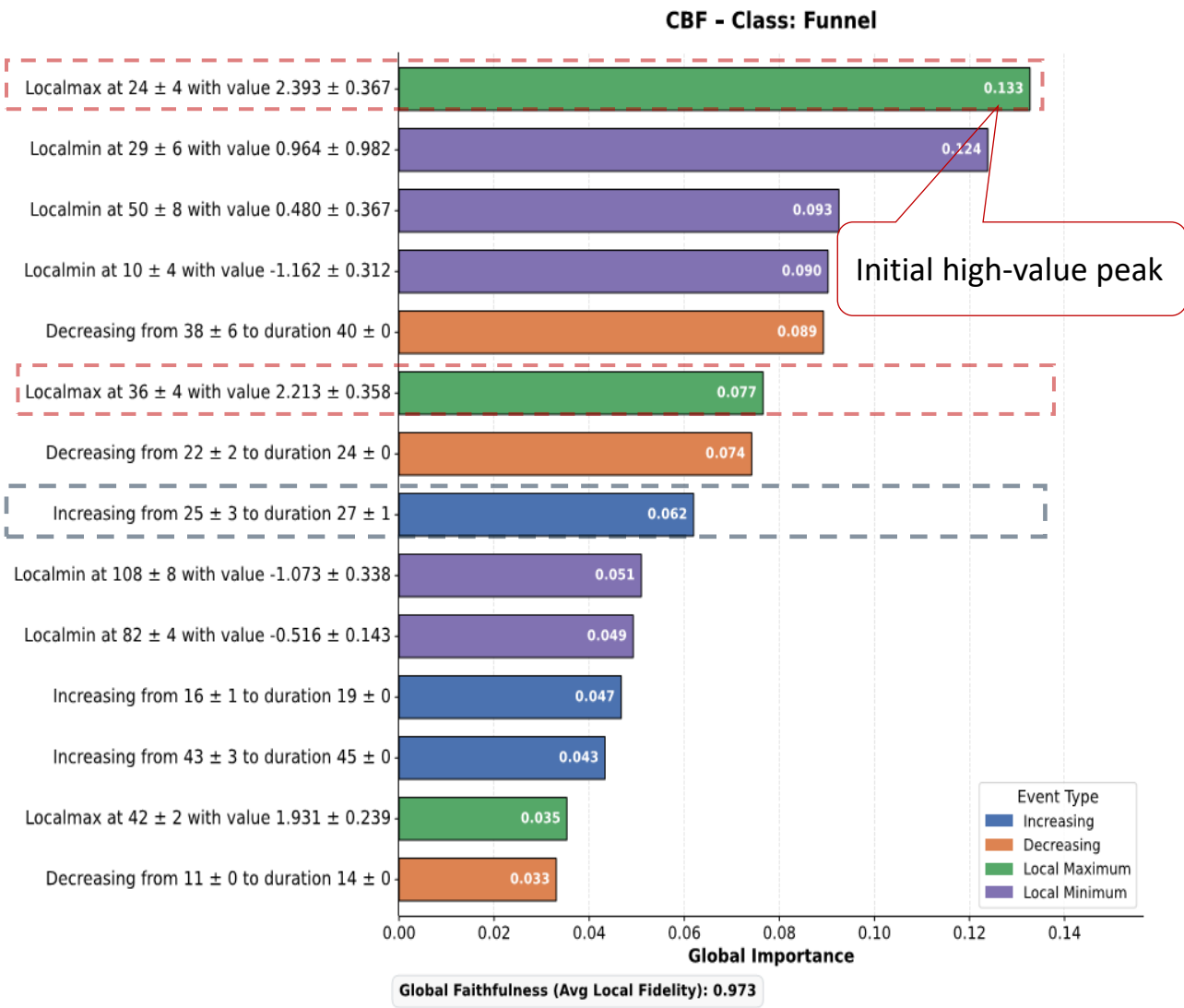
■ Both models identify the rise transition and neighbouring peaks.

CBF: L2GTX Global Explanations - Funnel

FCN

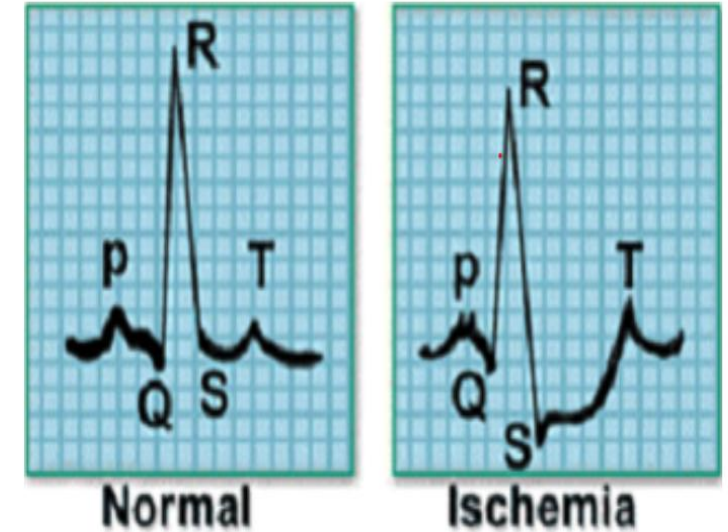
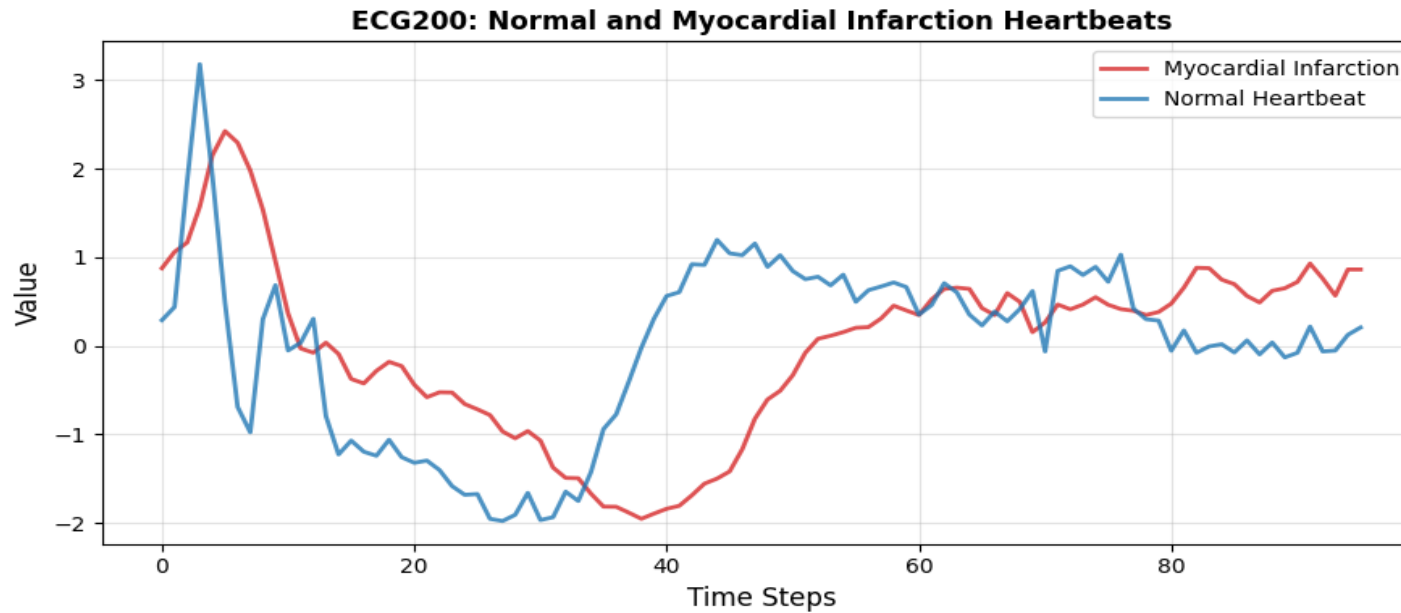


LSTM-FCN



- The most important events are concentrated around the rise transition, leading peak, and subsequent decline.

ECG200: Dataset Overview



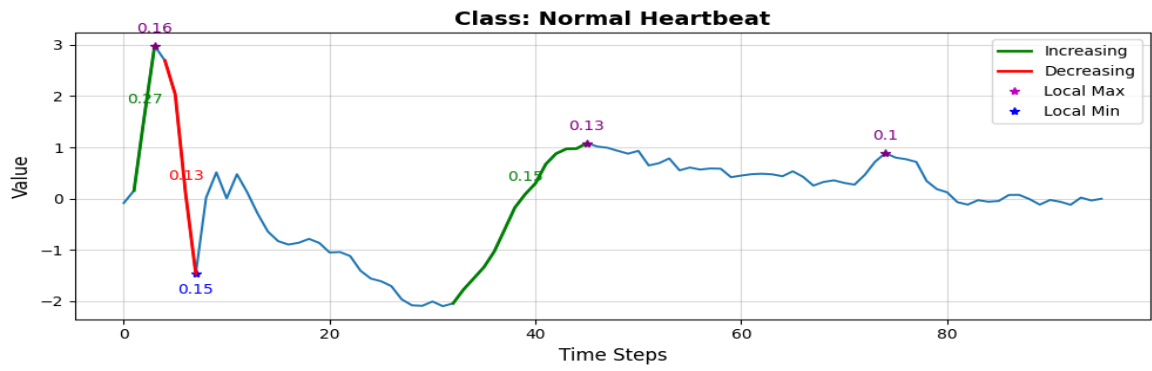
Clinical ECG illustration comparing normal and ischemic waveforms.
Source: UCR Time Series Classification Archive.

ECG200

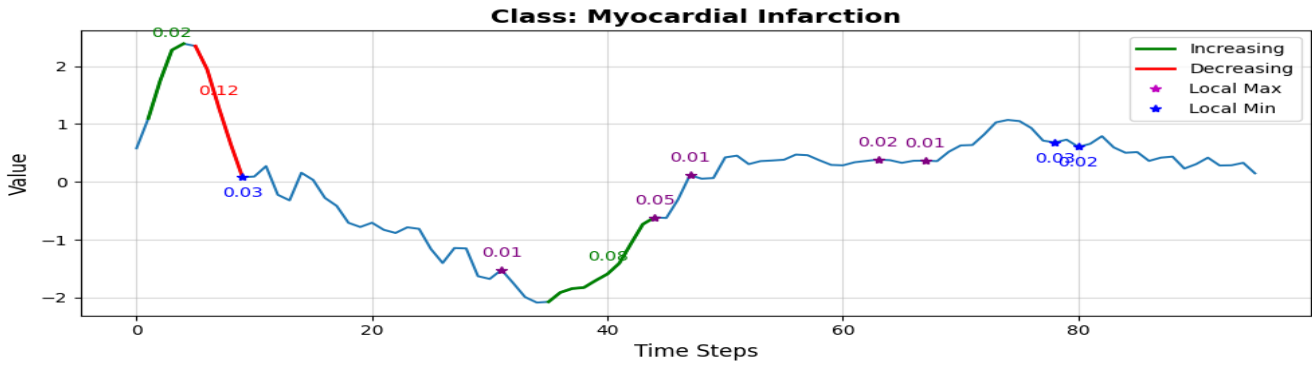
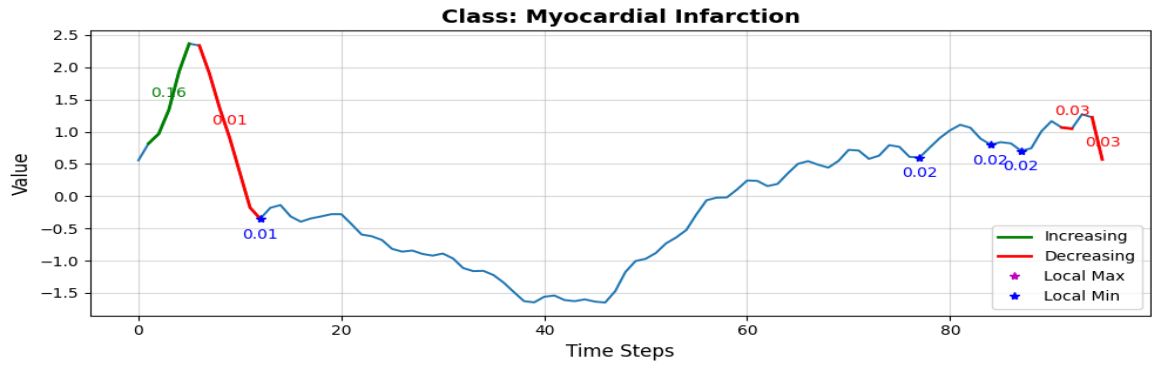
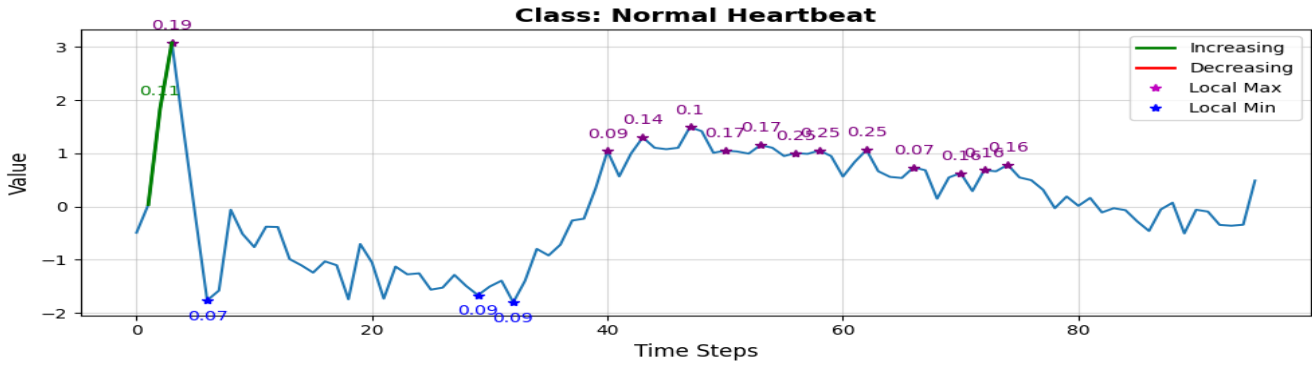
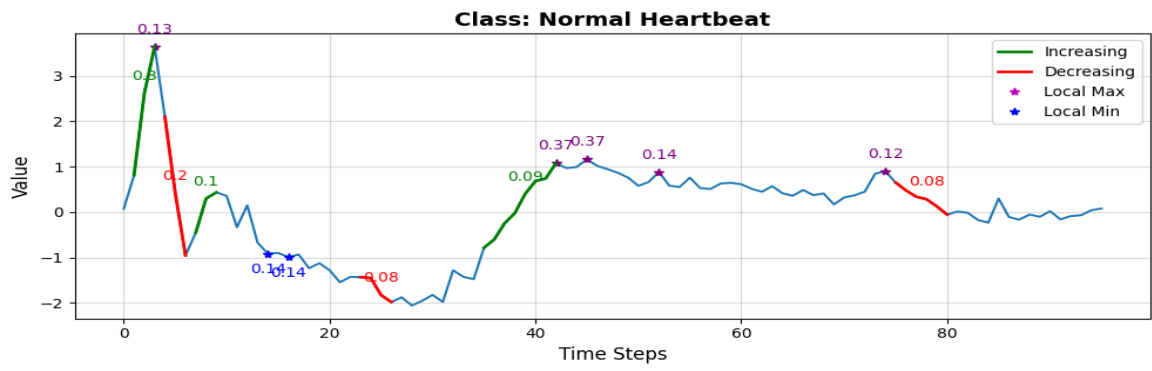
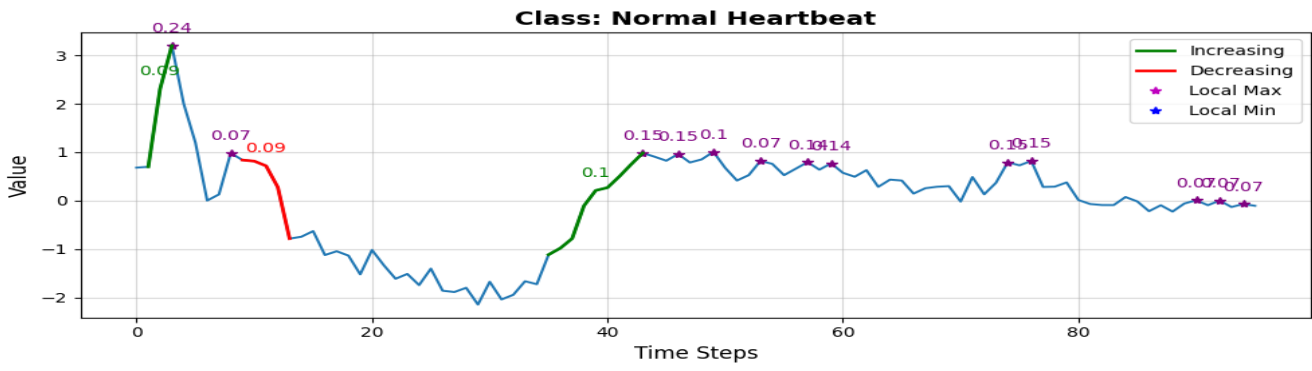
- Real-world ECG classification dataset.
- Distinguishes Normal Heartbeats and Myocardial Infarction (MI) heartbeats.
- Used to assess whether explanations capture clinically meaningful waveform differences.

ECG200: Local Explanations

FCN

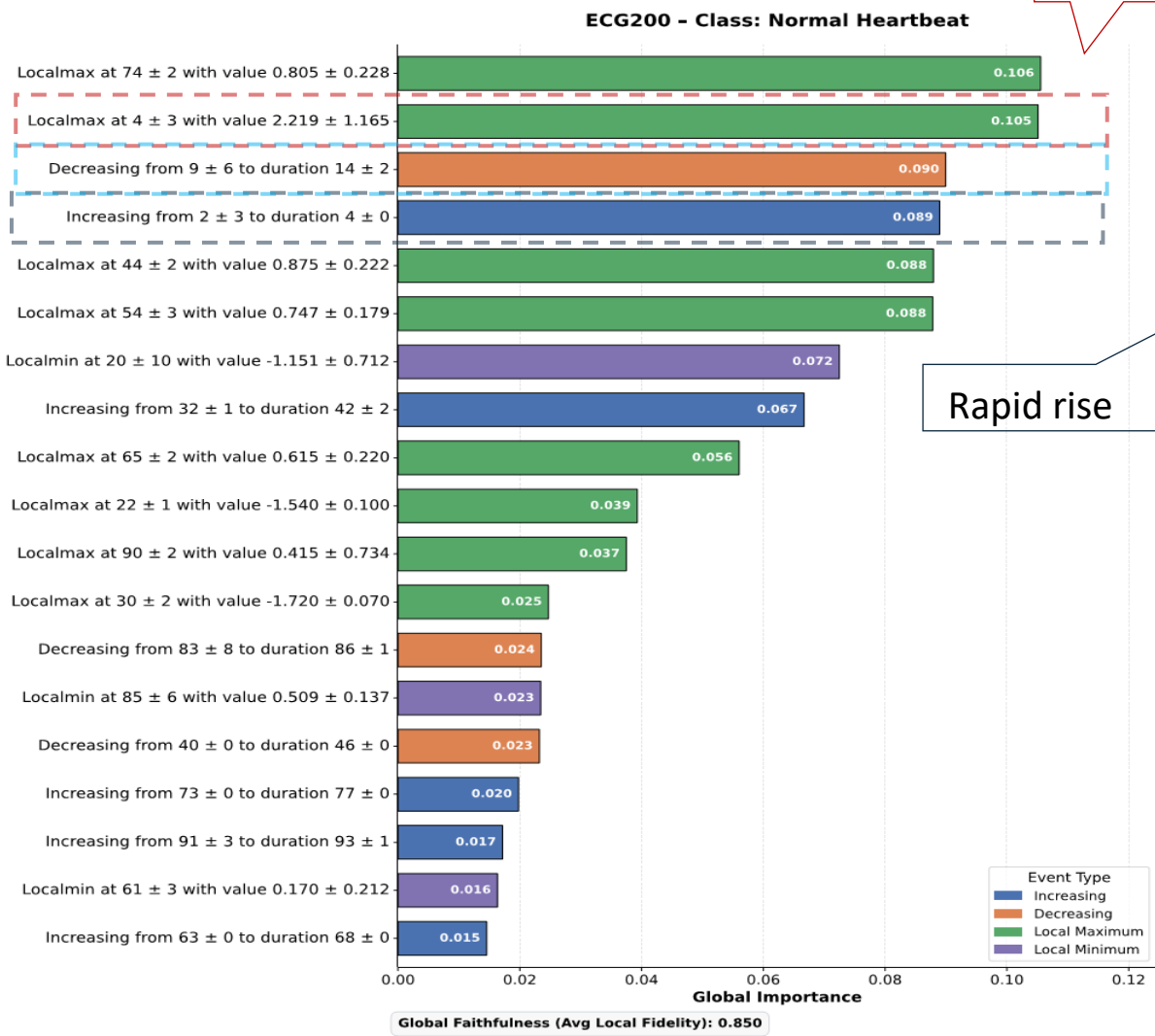


LSTM-FCN

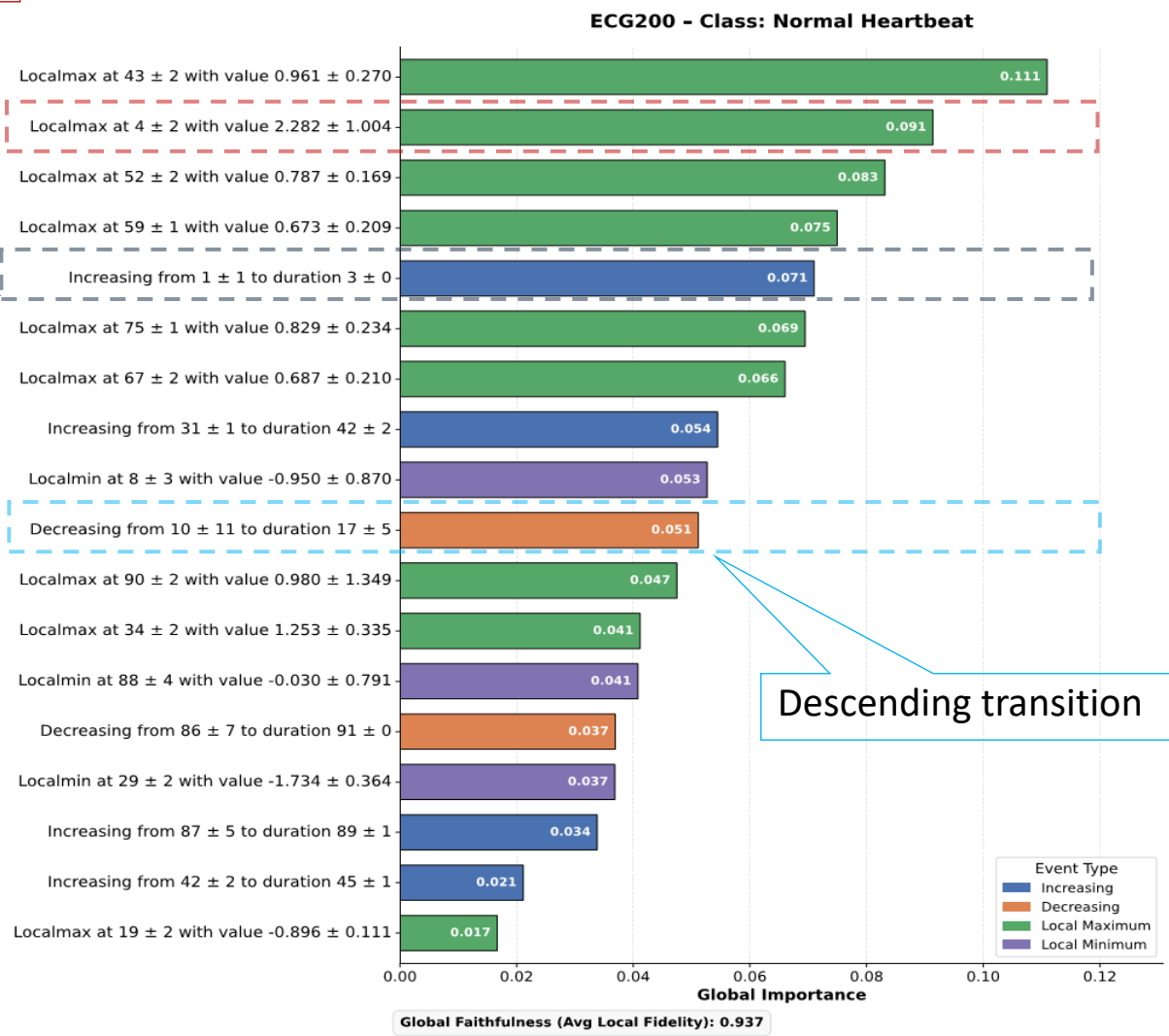


ECG200: L2GTX Global Explanations - Normal Heartbeat

FCN



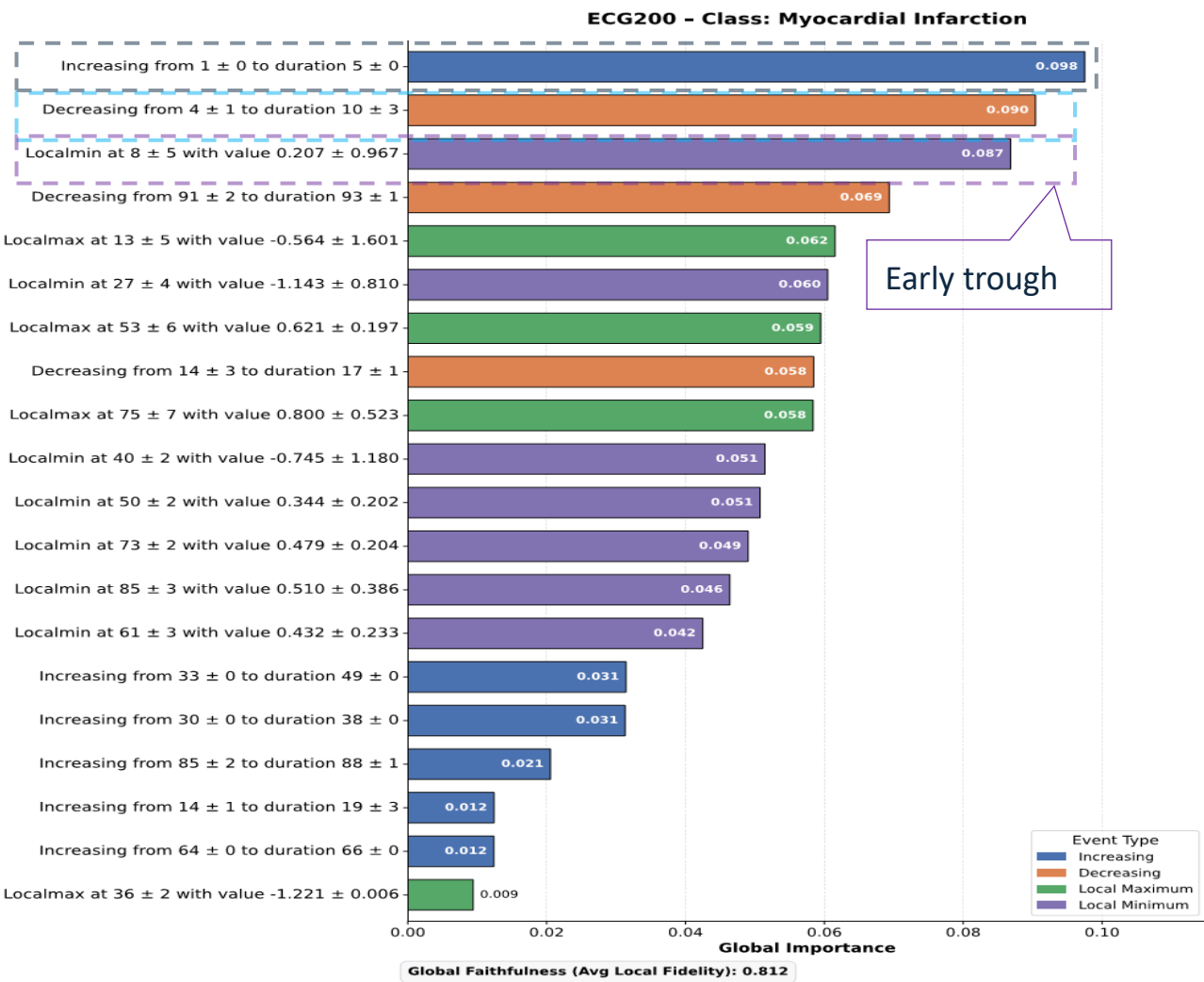
LSTM-FCN



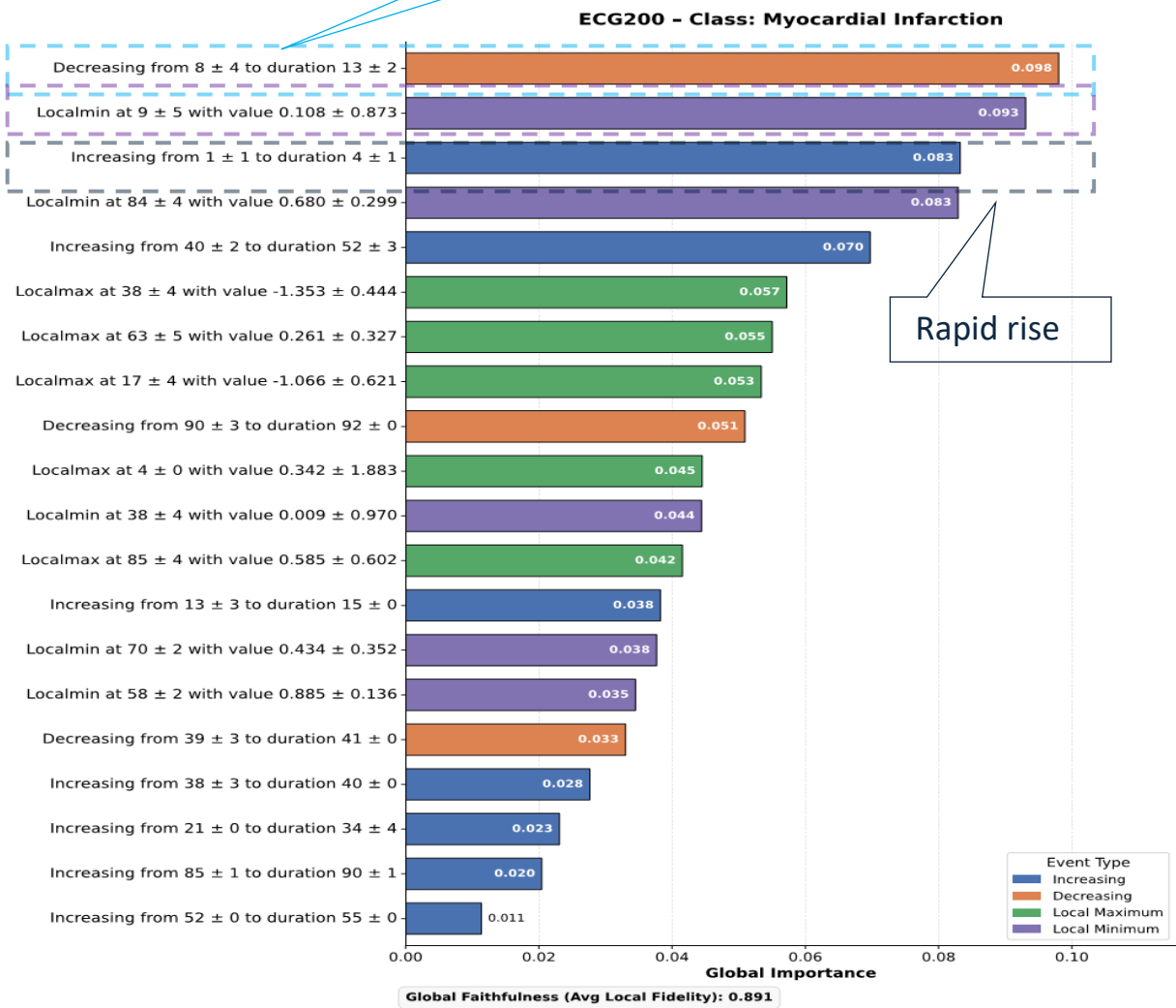
- The most important events capture the dominant initial peak and surrounding transitions.

ECG200: L2GTX Global Explanations - Myocardial Infarction

FCN



LSTM-FCN



■ The most important events are concentrated around the rapid rise, sharp decline, and early trough.

Contribution & Key Takeaway

Main Contribution

- L2GTX: A model-agnostic local-to-global framework for generating class-wise global explanations of time-series classifiers.

Key Characteristics

- ✓ Class-wise explanations
- ✓ Temporal-aware
- ✓ Human-readable
- ✓ Model-agnostic

Key Takeaway

L2GTX enables model-agnostic synthesis of class-wise global explanations from local explanations while preserving temporal structure.

Future Directions

- Extend to multivariate time series data.
- Human-centred evaluation with domain experts.
- Improve scalability of event-level clustering for resource-constrained environments.

Thank You!

Questions?



Paper



Code



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